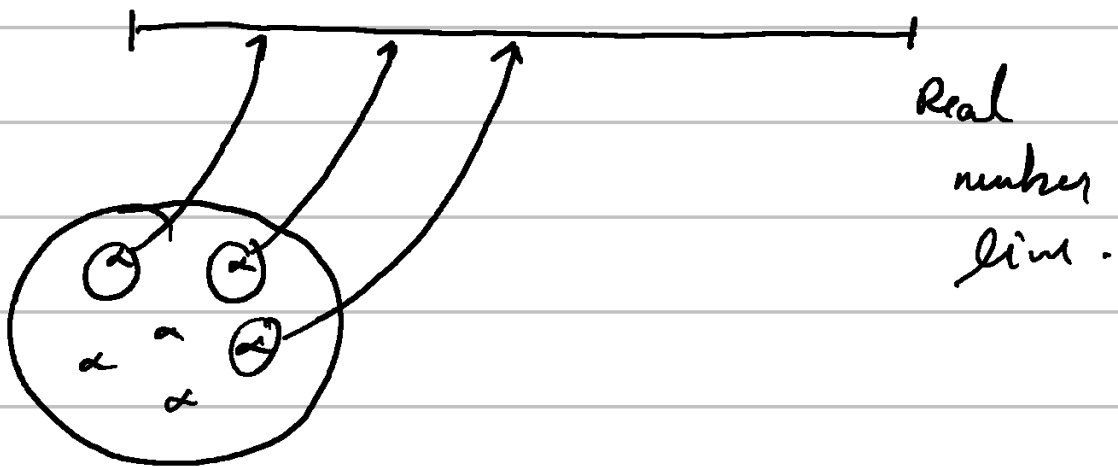


Discrete Random Variables

⇒ Given an experiment and the corresponding sample space, a r.v. associates a particular number with each outcome.



ex. roll of a dice.

$\{1, 2, 3, 4, 5, 6\}$

Main concepts

- ① A function of a r.v. is a r.v.
- ② We can associate with each r.v. certain "averages" of interest such as mean, variance. . . .
- ③ A r.v. can be conditioned on an event or another r.v.
- ④ A r.v. can be independent of another r.v.

R.v. is called discrete if its range is finite or at most countably infinite.

A discrete R.V. has an associated probability mass function.

Key term $\rightarrow$ Probability Mass Function (PMF)

$\hookrightarrow$ Gives the probability of each numerical value that the R.V. can take.

$P_X(a)$ $\hookrightarrow$ value of R.V.
 $\downarrow$
D.R.V. $X \in \Omega$ (sample space)

$$P_X(x) = P[X=x]$$

Normality axiom

$$\sum_{x \in \Omega} P_X(x) = 1$$

Famous probability mass functions

① Bernoulli R.V.

$$X \in \{0, 1\}$$

$$P[X=1] = p$$

$$P[X=0] = 1-p$$

② Binomial R.V.

A coin is tossed n times. At each time the head comes with proba. p .

Let X be number of heads in n tosses.

$X \rightarrow$ binomial random variable. with n, p

$$X = \{0, 1, 2, \dots, n\}$$

$$P_x(k) = P[X=k] = {}^n C_k p^k (1-p)^{n-k}$$

③ Geometric R.V.

It is the number of tosses needed for head to come up for the first time.

$$X = \{1, 2, 3, \dots, \infty\}$$

$$P_x(x) = P[X=x] = (1-p)^{x-1} \cdot p$$

④ Poisson R.V.

$$\Omega = \{0, 1, 2, \dots, \infty\}$$

$$P_x(k) = P[X=k] = \frac{\lambda^k}{k!} e^{-\lambda}$$

where $\lambda > 0$

⇒ Feel for poisson r.v. → binomial with very small p & very large n .

⇒ Poisson PMF with parameter λ is a good approximation for binomial PMF.

$$e^{-\lambda} \frac{\lambda^k}{k!} \approx \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

for $\lambda = np$, n large, p very small.

Functions of R.V.

Given a P.R.V. X , one may define other R.V. by applying various transformations on X .

X : Temperature in $^{\circ}\text{C}$

Y : $1.8X + 32$ in $^{\circ}\text{F}$

$$Y = g(X)$$

$$P_Y(y) = \sum_{\{x/g(x)=y\}} P_X(x)$$



Summation for cases when mapping is not one-to-one

Expectations, Mean, Variance

- Expectation of X is weighted average of all values of X .
→ Used to "summarize" the information, behaviour of X .

eg Money game based on a roll of a die.

→ Value of die gives money you win.

$$\rightarrow \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 \dots + \frac{1}{6} \cdot 6$$

$$= \frac{1}{6} \cdot 21 = \frac{21}{6} = \frac{7}{2}$$

3.5
expected winnings

$$E[X] = \sum_{x \in \Omega} P_X(x) \cdot x$$

- Mean is a representative value of X , also the expectation. gotcha!

Variance, Moments

- We define the 2nd order moment of a R.V. X as the expectation of X^2 .
- More generally: the n^{th} order moment of $X \rightarrow E[X^n]$
- An important quantity associated with X is its variance.

$$\begin{aligned}
 \text{VAR}[x] &= E[(x - E[x])^2] \\
 &= E[x^2 - 2xE[x] + E[x]^2] \\
 &= E[x^2] - 2E[x]^2 + E[x]^2 \\
 &= E[x^2] - E[x]^2
 \end{aligned}$$

variance is always non-negative.

$$\text{VAR}[x] = \sigma_x^2$$

→ $\sigma_x = \sqrt{\text{VAR}[x]}$: standard deviation

$\text{VAR}[x]$ provides a measure of dispersion around its mean.

Expected Value Rule

Let X be a R.V. with pmf $p_x(x)$
& let $g(x)$ be a fct of x .

$$E[g(x)] = \sum_{x \in \Omega} g(x) \cdot p_x(x)$$

eg $Y = aX + b$

$$E[X] = m_x$$

$$\text{VAR}[X] = \sigma_x^2$$

$$E[Y] = ?$$

$$\Rightarrow E[Y] = \sum_{x \in \Omega} (ax + b) p_x(x)$$

$$= \sum a x p_x(x) + \sum b p_x(x)$$

$$= \underline{\underline{a E[X] + b}} = \underline{\underline{a m_x + b}}$$

$$\Rightarrow \text{VAR}[Y] = E[Y^2] - (E[Y])^2$$

$\hookrightarrow am_x + b$

$$E[Y^2] = E[(aX + b)^2]$$

$$= E[a^2X^2 + b^2 + 2abX]$$

$$\Rightarrow a^2 E[X^2] + b^2 + 2abm_x - (am_x + b)^2$$

$$= a^2 E[X^2] + \cancel{2abm_x} + \cancel{b^2} - a^2m_x^2 - \cancel{b^2} - \cancel{2abm_x}$$

$$= a^2 (E[X^2] - E[X]^2) = a^2 \sigma_x^2$$

Exercise

X is a D.R.V.

$$\Omega_x = \{1, 2, 3, \dots, \infty\}$$

$$P_X(k) = P[X=k] = \frac{\alpha}{k!}$$

- i) Compute α
- ii) Compute $E[X]$
- iii) Compute $E[X(X-1)]$
- iv) Reduce $\text{VAR}[X]$.

$$\underline{\underline{i)}} \quad \sum_{k=0}^{\infty} P_X(k) = 1$$

$$\frac{\alpha}{1!} + \frac{\alpha}{2!} + \frac{\alpha}{3!} + \dots = 1$$

$$\alpha \left[\sum_{n \in \mathbb{N}} \frac{1}{n!} + 1 - 1 \right] = 1$$

e

$$\alpha = \frac{1}{e-1}$$

$$ii) \quad E[X] = \sum_{k=1}^{\infty} k \cdot \frac{\alpha}{k!} = \sum_{k=1}^{\infty} \frac{\alpha}{(k-1)!}$$

$$= \alpha e = \underline{\underline{\frac{e}{e-1}}}$$

$$\text{iii)} \quad E[X(X-1)] \quad \frac{\infty}{0}$$

$$Y = X(X-1)$$

$$\begin{aligned} E[X(X-1)] &= \sum_{k=1}^{\infty} k(k-1) \cdot \frac{\lambda}{k!} \\ &= \sum_{k=2}^{\infty} \frac{\lambda}{(k-2)!} = \frac{e}{e-1} \end{aligned}$$

$$E[X(X-1)] = \frac{e}{e-1}$$

$$E[X^2] - E[X] = \frac{e}{e-1}$$

$$E[X^2] = \frac{2e}{e-1}$$

$$\text{iv) } \sigma_x^2 = E[x^2] - (E[x])^2$$

$$= \frac{2e}{e-1} - \left(\frac{e}{e-1} \right)^2$$

$$= \frac{2e(e-1)}{(e-1)^2} - \frac{e^2}{(e-1)^2}$$

$$= \frac{2e^2 - 2e - e^2}{(e-1)^2}$$

$$= \frac{e^2 - 2e}{(e-1)^2} = \frac{e}{e-1} \cdot \frac{e-2}{e-1}$$

